Team — Team ShasthoSathi

- Rudra Sarker — BSc Industrial & Production Engineering, SUST (2027) · AI/ML builder
- Built & shipped: MindWell (mental-health web app), RippleUp (Android + Desktop), SightlineAI, Team SignTalk — plus this project, already live
- Advisor slot: public-health / field-implementation mentor (to be confirmed in Ignition Chamber)
- Team of one (eligible: 1–3 members) · Contact: via the FutureMakers portal

ShasthoSathi

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Everything demonstrated today is deployed and clickable — not a concept drawing.

scan → open → airplane-mode test

Problem — dengue outruns the first line of care

321,179	1,705	41,032 / 113	16,312
hospitalized (2023 — worst ever, DGHS)	deaths in 2023 (CFR 0.53%)	2026 cases/deaths to 5 Sep (DGHS)	of them in Dhaka division alone

- Community health workers (CHWs) meet every patient first — with paper, memory, and no AI support
- Internet is unreliable where they work → any cloud-only app fails exactly when needed
- Outbreak detection lags weeks; resources (test kits, volunteers, beds) are allocated without current division-level evidence
- Feature-phone and low-literacy users are excluded by app-first solutions

Target group — who we serve, first

- PRIMARY USERS — community health workers & supervisors in dengue-endemic wards (urban + rural)
- PRIMARY BENEFICIARIES — low-income families at first contact: low-literacy users, feature-phone users (SMS mode), elderly patients
- INSTITUTIONAL — city corporations, DGHS/IEDCR programme officers, NGOs needing early-warning + prioritization views
- Designed WITH the constraint, not against it: voice-first Bangla UX, offline-first architecture, no smartphone required for SMS mode

Evidence — 64 verified papers + live government data

- Peer-reviewed foundation: 64 Crossref-verified papers — Q1 spotlight incl. The Lancet Global Health (2024 dengue vaccine trial), Nature Medicine, PLOS NTD, IJID
- Clinical core: WHO 2009 dengue classification (warning signs validated — Hadinegoro 2012)
- Data spine: DGHS dashboard (2023→2026 YTD, division-level), WHO, IEDCR, BBS Census 2022, Open-Meteo weather — every number traceable, zero synthetic data
- Model honesty, in public: R^2(log)=0.93 on 2023; live 2026 check shows shape captured (Aug = actual peak), amplitude gap ~2.8× disclosed in-app + Model Card
- Automated data-integrity tests: build FAILS if any dataset stops matching its published totals

Solution — ShasthoSathi (live today)

- 🩺 Symptom triage — speak symptoms in Bangla → WHO-2009-based engine answers: home care / hospital today / EMERGENCY — with the triggering signs shown
- 💬 SMS mode — the SAME engine answers coded SMS on feature phones (GSM-7/UCS-2 aware)
- 🤰 Mother & child — WHO 8-contact ANC + Bangladesh EPI vaccine schedules + SMS reminders; 💊 medicine label read-aloud (OCR); 🎓 warning-signs quiz
- 🧠 Supervisor dashboard — 2023→2026 official data, real division map (Aug 2026: 20,536 cases — the peak month), weekly bulletin generator
- ⚡ Offline-first PWA: installable, free, no API keys — verified working in airplane mode

Role of AI — meaningful, explainable, responsible

AI component	Input → Output	Why AI / guardrail
Bangla speech recognition	voice → symptom list	removes literacy barrier
Explainable triage engine	flags → care level + reasons	WHO rules as auditable code — not a black box
SMS NLP	coded text → same triage	reaches feature phones
Climate-lag ML model	rain/humidity → risk index	weather leads cases by weeks; limits disclosed
On-device OCR + TTS	medicine photo → read-aloud	independent use by low-literacy patients

Responsible-AI position: AI is decision SUPPORT — every screen says “not a doctor” · personal data stays on-device · model limits published in-app (Model Card).

User journey — 90 seconds in a CHW's day

- 1️⃣ Open ShasthoSathi (home screen — works offline) → 2️⃣ Speak symptoms in Bangla
- 3️⃣ Engine answers: “hospital TODAY” + shows which WHO warning signs triggered
- 4️⃣ One-tap call 999/16263 · follow-up auto-created for tomorrow · case saved on-device
- 5️⃣ Supervisor dashboard: new ward cases appear in the division map → bulletin drafted → kits/volunteers prioritized
- 6️⃣ Feature-phone path: SMS “DENGUE FEVER PAIN” → same reply logic (simulated in-app; gateway-ready)

Expected impact — faster decisions, wider inclusion

- Clinical: decisions in the critical window (fever days 3–7, WHO) become consistent, explained, and escalation-biased — fewer missed warning signs
- Inclusion: voice-first Bangla + SMS reaches users that app-first tools exclude — measured by design, not by retrofit
- System: supervisors allocate from CURRENT division data (2026 live) instead of weeks-old reports
- Scalable public good: clinical content + language packs are data files — any dengue-endemic country adopts with two JSON edits
- Measurable pilot metrics: triage-to-referral time, warning-sign capture rate, follow-up compliance %, dashboard adoption by supervisors

Feasibility — it already runs; the hard part is done

- Built on free, proven pieces: no-build PWA (service worker), on-device speech/OCR, Leaflet offline maps, public data (DGHS/WHO/Open-Meteo/geoBoundaries/BBS)
- Zero running cost today: no servers required for field use (offline), no API keys, GitHub Pages hosting
- Quality gates: 55 automated tests (triage logic, schedules, i18n, SMS segmentation, data-integrity guards) · browser-verified offline mode
- Evidence discipline: 64 verified papers; automated guard tests fail the build on any data drift
- Risks owned: clinical misuse → guardrails + hotlines; privacy → on-device by design; model overconfidence → disclosed limits + relative-index outputs

Financial considerations — lean by architecture

- Now: BDT 0/month — offline-first removes server costs; hosting free (GitHub Pages); models public/free
- Pilot (12 months, est.): SMS gateway volume for reminders/alerts + field logistics + a domain — the dominant cost is coordination, not technology (est. figures in pilot plan; no proprietary pricing assumed)
- Unit economics: one CHW phone already in hand → marginal cost per additional user ≈ BDT 0
- Sustainability: ward/city-corporation or NGO sponsorship of SMS volume; zero-rating on health-education traffic is a natural telecom partnership (aligned with operator social-good programmes)
- No revenue claim: this is a public-interest tool; any future service fees would be disclosed and never charged to patients

Implementation — 12-month plan

- Months 0–1: pilot with one ward CHW team (Sylhet region) — baseline vs ShasthoSathi on triage consistency & referral time
- Months 2–4: case-feedback forecasting (needs DGHS/IEDCR division-monthly data partnership) — fixes the disclosed amplitude gap
- Months 3–6: SMS gateway delivery (simulation already proves format); supervisor sync portal
- Months 6–12: second city pilot + Bangla refinement from field transcripts; external clinical review of content
- Go-to-market: CHW programmes of DGHS/community-clinic network + NGO health programmes — we integrate into existing workflows, not replace them

Scalability & roadmap

- Bangladesh: ~169.8M population (BBS 2022); CHW networks among the world's largest — the constraint is coordination, and our unit economics make that the only constraint
- Global South: dengue-endemic countries (India, Indonesia, Brazil, Philippines) reuse the platform with two data files (clinical content + language) — architecture unchanged
- Roadmap: on-device Gemma-class LLM for richer consultation summaries (RAM-permitting) · ward-level case-feedback model · vaccine/ANC SMS automation via gateway
- Built for GP's assets: SMS gateway (simulated in-product) → live for every feature-phone user · zero-rating health traffic → free for all GP customers · MyGP integration · GP-brokered DGHS data partnership closes the forecast gap
- What we ask from FutureMakers: mentorship for the DGHS data partnership, pilot-site introduction, and responsible-AI guidance
- The idea is live today: rudra496.github.io/shasthosathi — open it during Q&A, turn on airplane mode, and try to break it.